

MANOJ KAVIKONDALA

Senior Technical Lead | Automotive Embedded Software Testing & Validation

Kavikondalamananoj@gmail.com | +81 80-3005-1991 | Utsunomiya, Japan



EXECUTIVE SUMMARY

Automotive embedded software testing authority with 12+ years of end-to-end V-cycle expertise, spanning HIL commissioning, OTA validation, and safety-critical system verification for Tier-1 OEMs including Honda R&D, BMW, ZF, Nexteer, and Ford. Architect of Python-driven test automation frameworks that cut manual analysis overhead by 60%+. Proven track record of leading cross-functional validation programs for Active Safety, and OTA Update systems on dSPACE Scalexio and Vector CANoe platforms. ISTQB-certified, Scrum Master-certified, and actively upskilled in Adaptive AUTOSAR and Agentic AI combining deep technical mastery with the leadership gravitas to drive quality across global, multi-site programs.

CORE COMPETENCIES

HIL Testing & Commissioning dSPACE Scalexio, ControlDesk, AutomationDesk, ModelDesk, Synect, Test planning, strategy development, validation coordination across components and modules	OTA Systems Validation OTA Update Architecture, Software-Over-the-Air (SOTA), FOTA, UDS Diagnostics, Hardware-software integration testing, functional and system-level validation	Test Automation & Python Python (Data Analysis, Tooling), vTESTstudio, KITE, Automation Libraries, CI/CD automation pipelines, test framework architecture, automated regression testing
Vector Tools Ecosystem CANoe, CANape, Vector VT System, CAN/LIN/Ethernet protocol analysis	Requirements & ASPICE IBM DOORS (Classic/DNG), BEE CodeBeamer, Rational Change, V-Model lifecycle, Jira, Requirements analysis, requirements-driven test specification, test coverage mapping	Safety Domain ABS, YSC, Traction Control, EPS, Active Safety, Lighting Systems (OEM-level)

PROFESSIONAL EXPERIENCE

Sr. Test Leader | KPIT Technologies · Client: Honda R&D

Tokyo, Japan (Locally Recruited in Japan) Oct 2023 – Present (2 yrs)

- Managed end-to-end OTA update testing for Honda R&D, reviewing 1,000+ test cycles daily to find critical software issues and share clear reports, reducing defect escape by 40%.
- Built a Python-based data pipeline integrating Vector VT and KPIT KITE with CI/CD automation principles, automating 70% of manual work and cutting report time from 4 hours to under 45 minutes. Designed reusable test framework libraries supporting requirements-driven test specification and automated regression coverage improvements.
- Led automated testing for SOTA/FOTA scenarios, creating reusable test libraries that improved regression coverage by 35% each sprint.
- Worked closely with Honda R&D and KPIT teams to convert unclear requirements into clear, ASPICE-compliant test cases using IBM DOORS.

Technical Lead — HIL & Active Safety | BlueBinaries Engineering · Client: ZF TCI

Hyderabad, India (On-Site) · Nov 2022 – Oct 2023 (1 yr)

- Led HIL bench setup using dSPACE Scalexio, integrating 25 braking and chassis features (ABS, YSC, TC) and delivering the system 2 weeks ahead of schedule.
- Built vehicle dynamics simulation models in dSPACE ModelDesk and automated braking tests using ControlDesk, improving test repeatability by 50%.
- Managed hardware validation for 12+ ECU components, working with teams to fix 30+ critical issues before moving from SIL to HIL.

Test Coordinator — Body Electronics | Hinduja Tech Limited · Client: Brose India

Pune, India (On-Site) · Jul 2021 – Nov 2022 (1 yr 5 mos.)

- Engineered comprehensive test specifications in DOORS Classic and BEE CodeBeamer for Power-Operated Tailgate (POT) and Electronic Power Window systems, achieving 100% requirements traceability across V-model lifecycle phases.
- Orchestrated manual and CANoe/CANape-driven functional test campaigns, managing strategy, planning, and execution across 3 concurrent feature streams while maintaining zero critical escapes to customer.

Sr. Test Manager — EPS Systems | BlueBinaries Engineering · Client: Nexteer Automotive

Bengaluru, India (On-Site) · Jun 2020 – Mar 2021 (10 mos.)

- ▶ Developed rigorous test specifications in IBM DOORS DNG for Electronic Power Steering (EPS) systems, performing deep requirement analysis and establishing bidirectional traceability for 200+ EPS functional requirements.
- ▶ Executed HIL validation campaigns using dSPACE AutomationDesk and ControlDesk, verifying EPS torque-assist behavior under 50+ dynamic vehicle simulation scenarios.
- ▶ Implemented automated regression test sequences via dSPACE Synect, reducing regression cycle time by 45% and enabling continuous integration of EPS firmware updates.

Sr. Software Engineer — Verification & Diagnostics | Behr-Hella Thermocontrol (BHTC)

Pune, India (On-Site) · May 2017 – Jun 2020 (3 yrs 2 mos.)

- ▶ Planned and executed diagnostic verification testing against OEM UDS specifications using VW's VAS system (ODIS), validating HVAC ECU diagnostic services across BMW and VW platform programs.
- ▶ Engineered Python and dSPACE AutomationDesk automated test sequences, eliminating 60% of manual diagnostic validation effort and generating stakeholder-ready reports maintained in DOORS and SharePoint.
- ▶ **On-Site Validation, Germany (BMW & Opel Facilities):** Deployed on-site to BHTC's Germany facility to conduct real-vehicle HVAC system validation across BMW, Opel, and Bentley platforms executing end-to-end diagnostic and functional test cycles directly on production vehicles and securing OEM-accepted sign-off for HVAC ECU software across three distinct vehicle lines.

Software Engineer — Lighting Systems | KPIT Technologies · Client: FMPL Ford Motors

Chennai, India (On-Site) · Apr 2015 – May 2017 (2 yrs 2 mos.)

- ▶ Delivered functional verification for exterior lighting systems (head, brake, and reverse lights) across Ford vehicle platforms, ensuring 100% SRS requirement coverage through structured test design.
- ▶ Developed reusable Python-scripted AutomationDesk libraries for test automation, reducing test authoring time by 30% and establishing a reusable asset baseline adopted by 2 additional project teams.

Test Engineer — HIL & Active Safety | Techwagon Solutions Pvt. Ltd.

Bengaluru, India (Remote) · Mar 2013 – Apr 2015 (2 yrs 2 mos.)

- ▶ Commissioned HIL Test Bench infrastructure for fault injection and active safety simulation, executing ABS, Traction Control (TC), and Yaw Stability Control (YSC) validation scenarios establishing the test foundation for subsequent production sign-off.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Certification	Provider
ISTQB® Certified Tester Foundation Level (CTFL)	ISTQB®
Scrum Master Certified (SMC)	Project Management Institute
Lean Six Sigma Foundations	Project Management Institute
Automotive Over-the-Air Update (OTA)	Udemy
Generative AI & Agentic AI Practitioner	KPIT Academy
Generative AI for Data Analytics	Udemy

EDUCATION

Bachelor of Engineering — Electronics & Communications

JNTUH College of Engineering Hyderabad, India · 2008 – 2012 · Grade: A

LANGUAGES

Telugu Native	English Full Professional	Japanese Elementary	Hindi / Tamil Working
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